QUICK CARD · COMPOSITES LAB (KOMPOSITTLABBEN)

Stepcraft M700 – CNC mill

UCCNC · SolidWorks CAM · § = section in the user manual

Full user manual with photos: **scan the QR code** or go to **ravo10.github.io/USN-kongsberg-lab-mech**.

SAFETY

- **Do not use the machine** without training
- Clamps **≥ 50 mm** outside the milling area
- **Emergency stops** outside and inside the door
- Use at most **640 × 480 mm** of the table

A. Start-up §3.01

1. Turn on the **compressed air**.
2. Turn on the **vacuum** (switch above the mill).
3. Release the **emergency stops** (out and in).
4. Start **UCCNC** on the PC.
5. Turn on the **spindle** (flashes blue/red).
6. Click **Reset** bottom right.

B. Z zero point with the TLS §4.07

7. Clamp the stock.
8. **M6** → the tool used first.
9. Move above a **high point** of the stock.
10. Put the **TLS** on the stock, under the tool.
11. **Page Up/Down** to just above the TLS.
12. Press **TLS** in UCCNC and wait.

Each time the stock has a new thickness. X/Y: zero in UCCNC or use the probe.

Troubleshooting §3.07, §4.05

Wrong toolpath	Close and reopen UCCNC
Tool stuck	Chuck lock + spanner (§4.05)
Zero point lost	Find the origin with the 3D Touch Probe
Holes < Ø6 mm	Mill Ø2 mm, drill out by hand

F. Shutdown §3.08

- ☐ Program finished, spindle stopped
- ☐ Tool in the magazine (M6), home position
- ☐ Spindle off, UCCNC closed
- ☐ Emergency stops in (outside and inside)
- ☐ Vacuum and compressed air off
- ☐ Vacuum the table and the tool holder

C. Tools and magazine §4.01–4.05

- **DOUBLE HOME ALL** → **OFFSETS** to calibrate the tools.
- Tools **1–8**. No. 9 is disabled, no. **10 = 3D Touch Probe**. Change with **M6** + no.
- Tool 1 is the Z reference for the probe.
- Stuck tool: press in the **chuck lock**, turn with the spanner.
- Keep the chuck clean: **vacuum** it during service.

D. DXF → G-code in UCCNC §3.05

13. DXF layers: no. 1 = **0**, no. 2 = **11**, then **12, 13 ...** (clean up in QCAD).
14. **CAM** tab → **Import DXF File** → choose the layer.
15. Choose the **origin** and the **Operation type** (inside/outside).
16. **Generate Toolpath** → **Create G-code**, one layer at a time.

Example (Ø2 mm cutter)

Tool diameter	2
Final depth	-6
Cut per pass	2
Feedrate / plunge	235 / 80
Spindle speed	23000
Safe Z	10

E. For students (SolidWorks) §2

- Origin at the centre of a **circle or rectangle** (the probe can find it again).
- **Z up**, X/Y as the part lies in the machine.
- The part must fit inside **640 × 480 mm** minus the clamps.
- **Pocket** (Finish) recommended, with tabs if needed.
- Holes under Ø6 mm: separate sketch with **Ø2 mm** holes.